Ciem Cornelissen

PhD Researcher in Computer Science Engineering

ciemcornelissen.github.io | ciem.cornelissen@ugent.be | linkedin.com/in/ciem-cornelissen | github.com/ciemcornelissen

Google Scholar | ORCID: 0009-0003-0149-6722 | Ghent, Belgium

Education

Ghent University - imec — Ghent, Belgium
PhD in Computer Science Engineering — *Jun. 2024 – Present*

KU Leuven — Leuven, Belgium
Advanced Master of Artificial Intelligence – Magna Cum Laude — *Sep. 2022 – Jun. 2023*

Ghent University — Ghent, Belgium
Master of Science in Physics – Cum Laude — *Sep. 2020 – Jun. 2022*

Research Experience

PhD Researcher — Jun. 2024 – Present
IDLab, Ghent University - imec — *Ghent, Belgium*

- Developed Le MuMo JEPA, extending joint-embedding predictive architectures to multi-modal perception through learnable fusion tokens and achieving the strongest performance-efficiency trade-off among from-scratch multi-modal baselines (CVPR 2026 URVIS)
- Developed LaFreeWM, a label-efficient JEPA world model that infers discrete latent actions during action-label-free pretraining and uses 1% or 10% labelled trajectories for raw-action planning
- Developed LISA, a domain-adversarial deep learning framework for illumination-invariant multi-modal analysis, improving prediction generalization by over 20% across varying environmental conditions
- Created the OHSLIC algorithm for efficient on-device sensor segmentation, achieving 10x inference speedup (12.2ms vs 130.2ms) on an NVIDIA Jetson Nano for real-time embedded perception

Publications

LaFree World Model: Label-Efficient Latent Action Inference for JEPA World Models — 2026
- *Manuscript under review at NeurIPS 2026* (Sep. 2026)
- Infers discrete latent actions from unlabelled observations, then grounds the world model with limited labelled data for raw-action planning
- Authors: **C. Cornelissen**, S. Leroux, P. Simoens | Paper

Le MuMo JEPA: Multi-Modal Self-Supervised Representation Learning — 2026
- *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) – URVIS Workshop* (Jun. 2026)
- Self-supervised framework learning unified representations from RGB and companion modalities via learnable fusion tokens
- Authors: **C. Cornelissen**, S. Leroux, P. Simoens | Paper | Code

In-Field Mapping of Grape Yield and Quality with Illumination-Invariant Deep Learning — 2025
- *IEEE Internet of Things Journal* (Oct. 2025)
- Developed LISA framework for illumination-invariant hyperspectral imaging in precision viticulture
- Authors: **C. Cornelissen**, S. De Coninck, A. Willekens, S. Leroux, P. Simoens | Paper | Code

Adaptive Clustering for Efficient Phenotype Segmentation of UAV Hyperspectral Data — 2025
- *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) Workshops* (Mar. 2025)
- Introduced OHSLIC algorithm for computationally efficient, real-time phenotype segmentation
- Authors: **C. Cornelissen**, S. Leroux, P. Simoens | Paper | Code

Technical Skills

Research Interests: Self-Supervised Learning, Predictive Representations, World Models, JEPA, Multi-Modal Learning, Computer Vision
Perception: RGB/LiDAR/Thermal Fusion, Depth Estimation, Semantic Segmentation, 3D Object Detection, On-Device Inference
Programming & Tools: Python, PyTorch, C/C++, TensorFlow, Git, Docker, Linux, LaTeX
Languages: Dutch (Native), English (Fluent, C2 level)